

Sang-hwa Lee

✉ torytony24@gmail.com ☎ 010 4400 9422 @ [@torytony24](https://www.instagram.com/torytony24) github.io

Education

- Seoul National University**, B.S. in Electrical and Computer Engineering Mar 2023 – Present
- GPA: 3.9/4.3
 - Key Coursework: Graphics Programming, Linear System Theory, Machine Learning, Introduction to Robotics
- Korea Science Academy of KAIST**, High school diploma Mar 2020 – Feb 2023
- GPA: 4.1/4.3
 - Specialized in mathematics, physics, and computer science

Research Experiences

- 3D Vision Lab**, Seoul National University Jul 2026 – Aug 2026
- Summer vacation undergraduate intern paper study
 - Topics of motion imitation and motion generation in computer graphics (e.g. Deep-Mimic, AMP, SMP)
- Graphics and Media Lab**, Seoul National University Jul 2023 – Jun 2024
- Undergraduate Researcher*
- In-depth literature review and [ground-up implementation](#) of **Position Based Dynamics(PBD)** and its extension **XPBD** using C++ / OpenGL
 - Developed foundational research skills, including reviewing papers, seminar presentation, and version control using Git

Honors & Awards

- Presidential Science Scholarship** Mar 2023 - Present
- The most prestigious scholarship for undergraduate students majoring in science
- International Science and Engineering Fair (ISEF)**, Grand Award 3rd place May 2022
- Physics & Astronomy division
 - The largest existing international science research competition for middle and high school students

Projects

- Soap Film Physics-Based Simulation** [github repository](#)
- Real-time, interactive physics-based simulation framework for soap bubbles
 - Done as part of Graphics Programming final project
 - Proposed a novel method of realistic simulation using PBD
- Jell-Kkagi** [github repository](#)
- Pseudo-3D mini game built with Pygame
 - All computations including 3D rendering and soft-body simulation implemented from-scratch only using NumPy
- Universal VarNet for undersampled MRI reconstruction** [github repository](#)

- A robust undersampled MRI image reconstruction framework using ASPIN and model distillation, implemented with PyTorch (SSIM 0.9697)
- Done as part of SNU FastMRI Challenge

Skills

Languages: Korean(Native), English(Fluent)

Programming: Python, C++, MATLAB

Tools & Libraries: Git, LaTeX, PyTorch, OpenGL

Other Experiences

Enlisted personnel, Republic of Korea soldier assigned to the 8th US Army

Oct 2024 - Apr 2026

Korean Augmentation To the United States Army (KATUSA)

- Enhanced English communication and writing skills through daily tasks

Research Interests

Computer Graphics

- Physics-based simulation, differentiable physics
- Learning-based character control, motion imitation, motion generation

Embodied AI & Robotics

- Sim-to-real transfer, human-object interaction
- Reinforcement learning for control